

A2 Warm Up 11.1 and 11.2

Show your work, not just an answer :)

1. For a certain class, the following data table compares how many students have a cat or a dog. What is the probability that a student chosen randomly from the class has a cat and a dog?

	Has a cat	Does not have a cat
Has a dog	2	3
Does not have a dog	13	11

TOTAL 15 14 29

$$\frac{2}{29}$$

2. For a certain class, the following data table compares how many students play an instrument or a sport. What is the probability that a student chosen randomly from the class plays neither a sport nor an instrument?

	Plays an instrument	Does not play an instrument
Plays a sport	7	16
Does not play a sport	4	2

11 18 29

$$\frac{2}{29}$$

3. A committee must be formed with 2 teachers and 4 students. If there are 11 teachers to choose from, and 16 students, how many different ways could the committee be made?

$$\frac{11C_2 \cdot 16C_4}{1} = \frac{55 \cdot 1820}{1} = 100,100$$

KEY

4. What is the total number of different 13-letter arrangements that can be formed using the letters in the word INSPIRATIONAL?

3 i's
2 n's
2 a's

$$\frac{13!}{3! 2! 2!} =$$

$$259,459,200$$

5. A spinner is divided into five colored sections that are not of equal size: red, blue, green, yellow, and purple. The spinner is spun several times, and the results are recorded below:

Spinner Results

Color	Frequency
Red	7
Blue	16
Green	10
Yellow	9
Purple	5

TOTAL 47

Based on these results, express the probability that the next spin will land on red as a decimal to the nearest hundredth.

$$\frac{7}{47} = .1489$$

$$.15$$